

DIAL TOOL

Guidebook

beta 2.3 · 4 Oct 2026

DrawBot · macOS

github.com/duztman/dial-tool

A watch dial is a few things repeated round a centre. Dial Tool draws it that way: a list of **rings** – ticks, markers, numerals, bands – plus hands and a date window, in millimetres, as clean vectors for print, etching and laser.



FIELD WATCH

P. 7



DIVER

P. 7



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P. 8



PILOT, TYPE B

P. 9



CALIFORNIA

P. 10



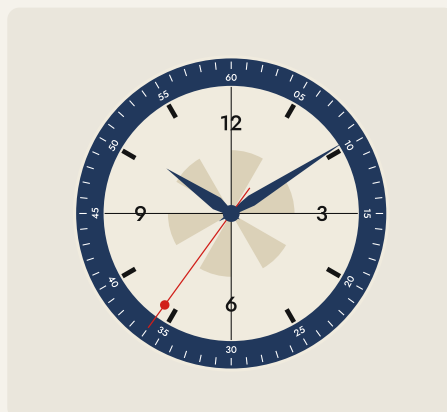
STATION CLOCK

P. 8



BAUHAUS

P. 9



SECTOR DIAL

P. 11



GAUGE

P. 10

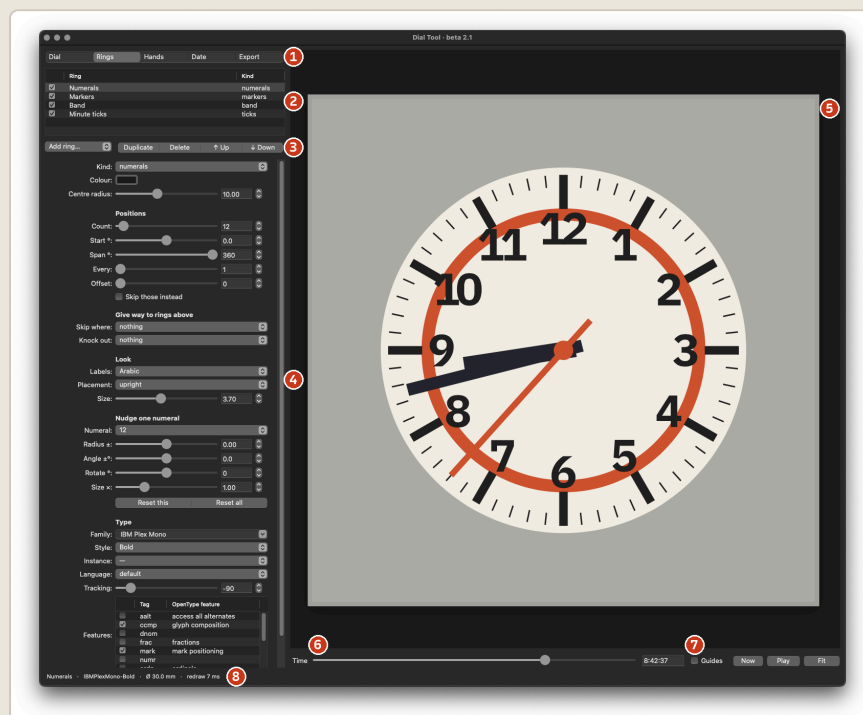
Every dial in this guide was drawn by the tool from the settings printed beside it. They are drawn by eye after well-known dial types – approximations, not measured copies. Rendered off-Mac with open fonts standing in for the macOS fonts named.

INSTALL

1. **DrawBot 3.132** or newer – free, macOS. [drawbot.com](https://drawbot.org/) → Download.
2. **dial.py** – github.com/duztman/dial-tool → **dial.py** → download the raw file. Branch **main** is released; **dev** is the newest work.
3. Open **dial.py** in DrawBot and press **⌘R**. The window opens.
4. For speed (optional): in DrawBot, Python → Install Python Packages → **skia-pathops** → Go.

GOOD TO KNOW

- **⌘R** again replaces the window and keeps your settings. Quitting DrawBot keeps them too.
- Everything is in **millimetres**, at true size.
- Angles are **clock angles**: 0° is 12 o'clock, growing clockwise. 90° is 3 o'clock.
- Everything is a **filled shape** – no strokes – so widths are exact in Illustrator.

THE WINDOW screenshot: beta 2.1 – the Canvas · PDF switch now sits beside Guides

- 1 **Sections** – Dial · Rings · Hands · Date · Export.
- 2 **Ring list** – the top ring is drawn in front. Tick to show; double-click to rename.
- 3 **Add ring...** · Duplicate · Delete · Up · Down.
- 4 **The selected ring's settings** – one scrolling list. Rows that don't apply hide.
- 5 **Preview** – live. The selected ring's guides are orange.
- 6 **Time** – drag, or type **10:09:36**. Now. Play.
- 7 **Guides** – construction circles, never exported.
- 8 **Status** – ring · font · diameter · build and frame time.

PREVIEW

Pinch	zoom about the cursor
⌘ + scroll	zoom about the cursor
Two fingers	pan
Fit	the whole page again
Canvas · PDF	Canvas is fast and sharp. PDF is DrawBot's own viewer: the reference. They should match.

TYPING NUMBERS

Every number has a slider, a field and a stepper. In the field:

- 1.2** sets 1.2
- +0.5** adds 0.5
- x0.5** halves
- /2** halves, too

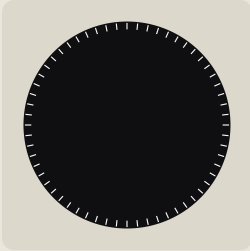
Typing past a slider's end extends the slider.

SECTIONS

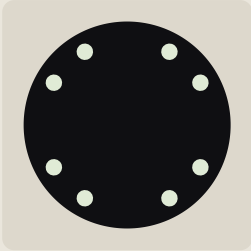
Dial	diameter, page margin, plate and backdrop colour
Rings	the list, and the selected ring – p. 3–5
Hands	hour, minute, seconds, cap – p. 6
Date	the window – p. 6
Export	files, settings, the log – p. 12

A ring repeats one thing round the centre, at its positions. The list order is the stacking order.

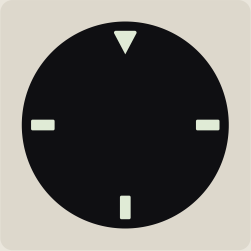
ONE DIAL, TAKEN APART the diver from p. 7 – five rings, top first



Minutes
ticks · Count 60



Dots
markers · Every 3, skip those instead



Bars · Triangle
markers · Count 4 · Count 1

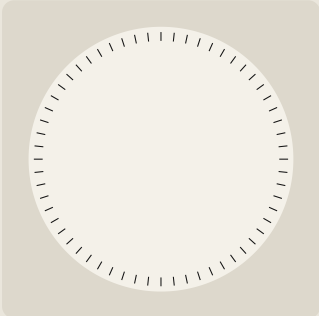


Depth
numerals · Custom · on path



= the dial
plus hands and the date

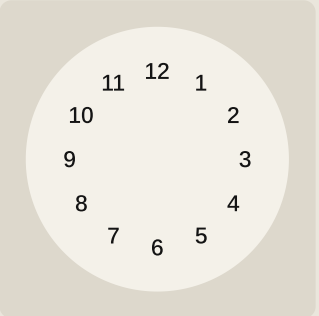
FOUR KINDS



ticks
Count 60 · Length 1.0 · Width 0.12



markers
Count 12 · Length 2.8 · Width 0.7



numerals
Labels Arabic · Size 2.6



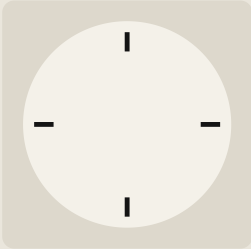
band
Radius 14.4 · Band width 2.2

Change a ring's **kind** at any time: every ring carries every setting, so nothing is lost. **Radius** is the outer edge of ticks, markers and bands; for numerals it is the centre of the text.

POSITIONS where a ring puts things



Count 12
Start 0° · Span 360°



Every 3
the cardinals



Every 3 · skip those instead
the other eight



Every 2 · Offset 1
the odd hours



Count 10 · Start -135° · Span 270°
an arc: both ends marked

Count	how many positions	Every	use every n-th position...
Start °	where the first one sits (0 = 12 o'clock)	Offset	...counting from this one
Span °	360 = all round, evenly. Less = an arc with both ends marked	Skip those instead	turn the choice round

IN THE LIST

Add ring... puts a new ring above the selected one.
Duplicate, Delete, ↑ Up, ↓ Down. The checkbox hides a ring without losing it. A new numerals ring copies the type of the one above.

COLOUR

Each ring has one flat colour. The plate and the backdrop are in the Dial section; hands, cap and date print have their own. The backdrop is for the preview only.

03 MARKS & BANDS

Ticks and markers share one set of shapes. Bands are colour. Rings give way to the rings above them.

SHAPES ticks and markers • values in mm



bar
Length 3.4 • Width 1.2



wedge
Inner width $\times 0.35$



wedge
Inner width $\times 0$



rounded
Round outer 0.6 • inner 0.2



dot
 $\varnothing 2.0$



hollow
Hollow wall 0.22



file
your SVG • PDF • AI

12 O'CLOCK a markers ring's first position



same



double
Gap 0.4



triangle
Scale $\times 1.2$



none

READING THE SETTINGS

Length • Width along the radius • across it

Inner width $\times$ 0 triangle • 1 bar • > 1 wider inside

Round true fillets, at either end

Hollow wall an outline of that thickness

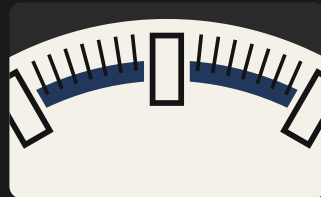
GIVE WAY TO RINGS ABOVE one mechanism for every pair of rings



stack
Skip where: nothing • Knock out: nothing



skip where
markers above



knock out
markers above • Clearance 0.25



knock out
markers & numerals above

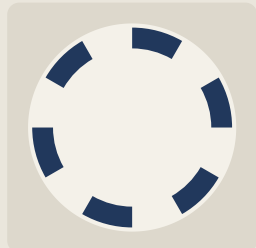
Skip where drops this ring's positions where the chosen rings above already have a mark: minute ticks that stop for the hour markers.

Knock out cuts a gap of **Clearance** mm round the shapes above — markers, markers & numerals, or everything. The date window always clears every ring.

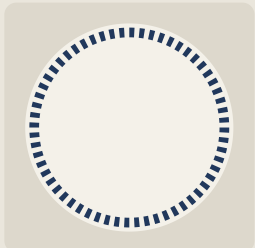
BANDS flat colour between Radius and Radius - Band width



ring
Count 1 • Band width 2.0



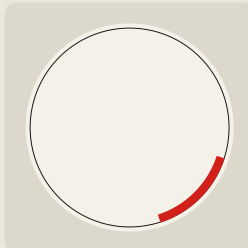
sectors
Count 12 • Every 2



fill
Count 60 • Fill 50 %



pies
Band width = radius



arc
Start 108° • Span 54°

For bands, each position starts a segment and the segments tile the span. **Fill %** is how much of its step a segment covers. Railroad rails are two thin bands; a sector dial is pies.

LABELS

11121

Arabic

XIXIXII I

Roman • IIII

١١ ٢١ ١

Eastern Arabic

+ − + = −

Chinese

ELEVEN TWELVE ONE

Words
on path

556005

Numbers
First 5 • Step 5 •
Digits 2 • Start
30°

NNW N NNE

Custom
N, NNE, , , E, ...

Also Roman • IV. Numbers count from First number by Step, padded to Digits. Custom is your own list, comma-separated; an empty entry leaves that position empty.

PLACEMENT

upright
always level

radial
turns with the circle

radial, auto-flip
never upside down

on path
letters follow the circle

On path sets each letter on the circle with the font's real spacing and one baseline; below 3 and 9 o'clock it flips to read left to right.

TYPE each numerals ring has its own – fonts as named on macOS

11121

DIN Alternate Bold

11121

Futura Medium

11121

Helvetica Neue Medium

XIXIXIXII I

Bodoni 72 Book

TYPE SETTINGS

the date uses the top numerals ring's

Family • Style

any font installed on the Mac. No fonts folder.

Instance

a variable font's named instances

Variable axes

one slider per axis the font has

Features

OpenType features, with the designer's own names: tabular or oldstyle figures, stylistic sets, slashed zero...

Language

local forms, e.g. Turkish İ ı

Size • Tracking

mm • 1/1000 em, like Illustrator

NUDGE

one numeral at a time

Pick a numeral, then correct it by eye. Stored per position; Reset this, Reset all.

Radius ±

in or out, mm

Angle ±°

round the dial

Rotate °

about its own centre

Size ×

larger or smaller

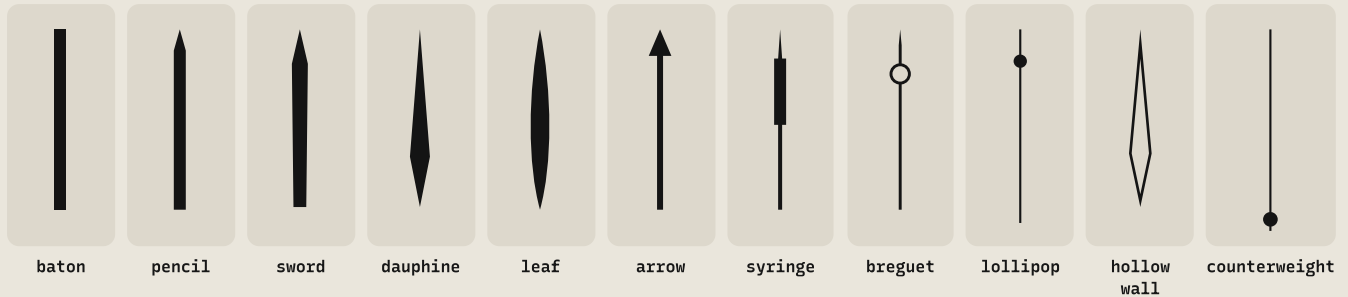
Typical:

pull a wide 12 in by -0.2, or shrink 10 • 11 • 12 with Size × 0.92 so two-digit numerals weigh the same as the others.

DIAL TOOL • GUIDEBOOK

BETA 2.3 • 05

SHAPES drawn here at Length 12



A HAND'S SETTINGS

Length · Width	from the pivot to the tip · at its widest
Tail	behind the pivot
Tip length	pencil, sword, arrow, syringe, Breguet
Head · ring · disc	arrow head or syringe barrel width; Breguet ring or lollipop disc Ø – and where it sits, in % of the length
Hollow wall	a lume opening
Counterweight Ø	a disc on the tail
File	your own drawing: Fit file to length, or Size ×; Width × stretches

MOTION the seconds hand

sweep	smooth
quartz · 1/s	one step a second
6 beats/s	21,600 vph
8 beats/s	28,800 vph

Play runs the hands from the set time; MP4 and GIF export record them. The **cap** over the pivot has its own diameter and colour.

DATE WINDOW



at 3
Radius 10.5 · 2.6 × 2.0 · Frame 0.15

at 4:30
Corner 0.6 · Frame 0

at 6
Print clearance 0.4 cuts the band

Position	3 · 4:30 · 6 · 9 · 12
Radius	the window's centre
Width · Height · Corner	the aperture

Frame	a printed border; 0 for none
Print clearance	every ring keeps this far from the hole
Shows · Text %	the day on the disc, and its size

Markers and numerals leave the date's position empty. In a production mask the aperture is a cut line. The frame and the day share the Print colour.



FIELD WATCH

after the A-11 and its descendants: 12 and 24 hours, a plain minute track, nothing else. [presets/field.json](#)

Dial	Diameter 30 · Plate ☒
Hours numerals	Radius 10.5 · Size 3 · Font DIN Alternate Bold ☐
24 hours numerals	Radius 7.3 · Labels Custom · Custom 24, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23 · Size 1.35 · Font DIN Alternate Bold ☐
Hour marks markers	Radius 14.4 · Length 1.7 · Width 0.42 ☐
Minutes ticks	Radius 14.4 · Count 60 · Length 1 · Width 0.16 · Skip where markers above ☐
Hour hand	Shape sword · Length 7.6 · Width 1.5 · Tail 1.6 · Tip length 2.4 ☐
Minute hand	Shape sword · Length 12.6 · Width 1.2 · Tail 1.8 · Tip length 3 ☐
Seconds hand	Shape baton · Length 13.2 · Width 0.14 · Tail 3.4 ☐
Cap	Ø 1.5 ☐



DIVER

after the 1950s dive watch: fat dots, bars at 6 and 9, a triangle at 12, the date at 3. [presets/diver.json](#)

Dial	Diameter 29 · Plate ☒
Triangle markers	Radius 13.2 · Count 1 · Shape wedge · Length 3.3 · Width 3.5 · Inner width × 0 · Round outer 0.25 ☐
Bars 6 · 9 markers	Radius 13.2 · Count 4 · Length 3.3 · Width 1.5 · Round outer 0.2 · Round inner 0.2 · 12 o'clock none ☐
Dots markers	Radius 13 · Every 3 · Skip those instead · Shape dot · Ø 2.3 ☐
Depth numerals	Radius 5.2 · Count 1 · Start 180° · Labels Custom · Custom 200 m · 660 ft · Placement on path · Size 0.95 · Font Helvetica Neue Medium ☐
Minutes ticks	Radius 14.3 · Count 60 · Length 0.9 · Width 0.16 ☐
Date window	Position 3 · Radius 10.7 · Width 2.7 · Height 2 · Corner 0.15 · Frame 0 · Print clearance 0.3 · Shows 24 · Text 68 %
Hour hand	Shape pencil · Length 7.4 · Width 2 · Tail 1.6 · Tip length 1.8 ☐
Minute hand	Shape pencil · Length 12.4 · Width 1.5 · Tail 1.8 · Tip length 2 ☐
Seconds hand	Shape lollipop · Length 13 · Width 0.18 · Tail 3.4 · Head · ring · disc 1.25 · at 70 % ☐
Cap	Ø 1.6 ☐



DRESS WATCH

after the Japanese dress watches of the 1960s: plain bars, a double bar at 12, dauphine hands, a framed date. [presets/dress.json](#)

Dial	Diameter 30.4 · Plate ☐
Hours markers	Radius 13.9 · Length 2.9 · Width 0.8 · 12 o'clock double · Gap 0.3 ☒
Minutes ticks	Radius 14.7 · Count 60 · Length 0.8 · Width 0.09 · Skip where markers above ☒
Line 1 numerals	Radius 5.6 · Count 1 · Start 180° · Labels Custom · Custom AUTOMATIC · Placement on path · Size 0.95 · Font Helvetica Neue Medium ☒
Line 2 numerals	Radius 7 · Count 1 · Start 180° · Labels Custom · Custom HI-BEAT 36000 · Placement on path · Size 0.7 · Font Helvetica Neue Medium ☒
Date window	Position 3 · Radius 10.9 · Width 2.7 · Height 2 · Corner 0.1 · Frame 0.22 · Print clearance 0.25 · Shows 8 · Text 70 %
Hour hand	Shape dauphine · Length 8.4 · Width 1.6 · Tail 1.4 · at 20 % ☒
Minute hand	Shape dauphine · Length 13 · Width 1.25 · Tail 1.6 · at 15 % ☒
Seconds hand	Shape baton · Length 13.9 · Width 0.12 · Tail 3.6 ☒
Cap	Ø 1.1 ☒



STATION CLOCK

after the Swiss railway clock of 1944: heavy bars, and a red seconds hand like a dispatcher's baton. [presets/railway.json](#)

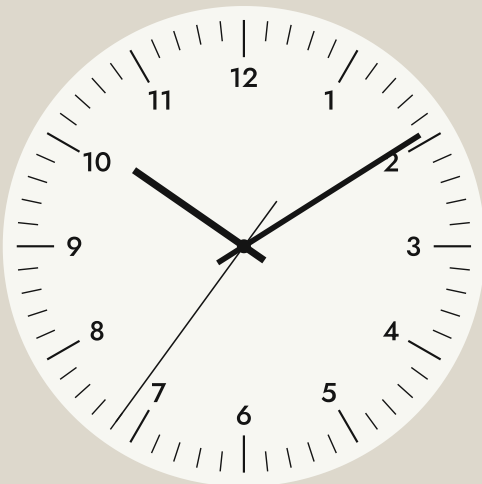
Dial	Diameter 30 · Plate ☐
Hours markers	Radius 14.3 · Length 3.7 · Width 1.3 ☒
Minutes ticks	Radius 14.3 · Count 60 · Length 1.3 · Width 0.45 · Skip where markers above ☒
Hour hand	Shape baton · Length 8.8 · Width 1.7 · Tail 2.4 ☒
Minute hand	Shape baton · Length 13.6 · Width 1.3 · Tail 2.8 ☒
Seconds hand	Shape lollipop · Length 9.8 · Width 0.34 · Tail 3.6 · Head · ring · disc 2.7 · at 100 % ☒
Cap	Ø 1 ☒



PILOT, TYPE B

after the 1940s observer's watch: minutes outside, hours on an inner ring, a triangle and two dots at 12. [presets/pilot.json](#)

Dial	Diameter 34 · Plate ●
Minutes numerals	Radius 12.6 · Count 11 · Start 30° · Span 300° · Labels Numbers · First number 5 · Step 5 · Digits 2 · Size 2.5 · Font DIN Alternate Bold ○
Triangle markers	Radius 14 · Count 1 · Shape wedge · Length 2.6 · Width 2.6 · Inner width × 0 ○
Two dots ticks	Radius 13.6 · Count 2 · Start -11° · Span 22° · Shape dot · Ø 0.7 ○
Five minutes markers	Radius 16.4 · Length 2.2 · Width 0.5 ○
Minute track ticks	Radius 16.4 · Count 60 · Length 1.3 · Width 0.2 · Skip where markers above ○
Hours numerals	Radius 6.6 · Size 1.7 · Font DIN Alternate Bold ○
Hour circle band	Radius 8.6 · Band width 0.14 ○
Hour hand	Shape sword · Length 7.4 · Width 2 · Tail 1.4 · Tip length 2.6 ○
Minute hand	Shape sword · Length 14.6 · Width 1.5 · Tail 1.8 · Tip length 3.6 ○
Seconds hand	Shape baton · Length 15.4 · Width 0.16 · Tail 4.2 · Counterweight Ø 1.1 ○
Cap	Ø 1.7 ○



BAUHAUS

after the German wall-clock school of the 1950s–60s: hairline marks, small numerals, thin hands. [presets/bauhaus.json](#)

Dial	Diameter 31 · Plate ○
Hours numerals	Radius 10.9 · Size 1.75 · Font Futura Medium ●
Hour lines markers	Radius 14.6 · Length 2.4 · Width 0.14 ●
Minutes ticks	Radius 14.6 · Count 60 · Length 1.3 · Width 0.09 · Skip where markers above ●
Hour hand	Shape baton · Length 8.6 · Width 0.46 · Tail 1.6 ●
Minute hand	Shape baton · Length 13.4 · Width 0.36 · Tail 2 ●
Seconds hand	Shape baton · Length 14 · Width 0.1 · Tail 3.6 ●
Cap	Ø 0.9 ●



CALIFORNIA

after the 1930s–40s dial: Roman on top, Arabic below, bars at 3 · 6 · 9, a triangle at 12, a railroad track. [presets/california.json](#)

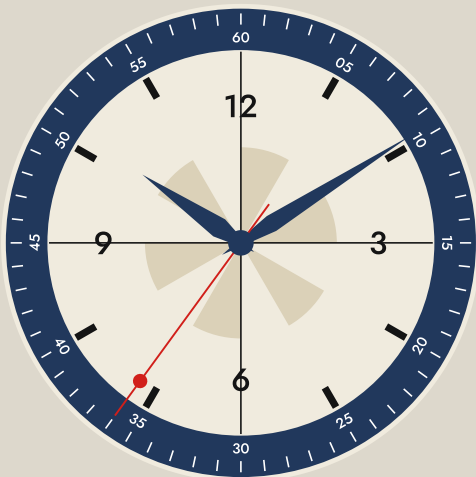
Dial	Diameter 30 · Plate ●
Roman numerals	Radius 9.6 · Labels Custom · Custom , I, II, , , , , , , X, XI · Placement radial · Size 3.3 · Font Bodoni 72 Book ○
Arabic numerals	Radius 9.6 · Labels Custom · Custom , , , , 4, 5, , 7, 8, , , · Placement upright · Size 3.2 · Font DIN Alternate Bold ○
Bars + triangle markers	Radius 11.4 · Count 4 · Length 3.4 · Width 1 · 12 o'clock triangle · Scale × 1.25 ○
Minutes ticks	Radius 14.2 · Count 60 · Length 1 · Width 0.14 ○
Outer rail band	Radius 14.2 · Band width 0.12 ○
Inner rail band	Radius 13.32 · Band width 0.12 ○
Hour hand	Shape pencil · Length 7.2 · Width 1.5 · Tail 1.4 · Tip length 1.6 ○
Minute hand	Shape pencil · Length 11.8 · Width 1.2 · Tail 1.6 · Tip length 1.8 ○
Seconds hand	Shape baton · Length 12.6 · Width 0.13 · Tail 3.2 · Counterweight Ø 0.9 ○
Cap	Ø 1.4 ○



GAUGE

a 270° scale with a red zone: positions over a partial span mark both ends. [presets/gauge.json](#)

Dial	Diameter 30 · Plate ○
Scale numerals	Radius 11.3 · Count 10 · Start −135° · Span 270° · Labels Numbers · First number 0 · Step 10 · Placement radial, auto-flip · Size 1.5 · Font Helvetica Neue Medium ●
Scale marks markers	Radius 14.3 · Count 10 · Start −135° · Span 270° · Shape wedge · Length 1.6 · Width 0.34 · Inner width × 0.3 ●
Scale ticks ticks	Radius 14.3 · Count 46 · Start −135° · Span 270° · Length 0.8 · Width 0.1 · Skip where markers above ●
Rail band	Radius 14.5 · Count 1 · Start −135° · Span 270° · Band width 0.12 ●
Red zone band	Radius 14.5 · Count 1 · Start 108° · Span 27° · Band width 0.9 · Knock out everything above · Clearance 0.12 ●
Minute hand	Shape arrow · Length 12.6 · Width 0.4 · Tail 2.6 · Tip length 2.2 · Head · ring · disc 1.5 ●
Cap	Ø 1.2 ●



SECTOR DIAL

after the 1930s scientific dials: a chapter ring, a crosshair, zones of colour. [presets/sector.json](#)

Dial	Diameter 30 · Plate ☐
Hours numerals	Radius 8.6 · Every 3 · Size 2 · Font Futura Medium ☒
Hour bars markers	Radius 11.9 · Every 3 · Skip those instead · Length 1.4 · Width 0.45 ☒
Minutes ticks	Radius 14.4 · Count 60 · Length 0.7 · Width 0.1 ☐
Minute numerals numerals	Radius 12.9 · Start 30° · Labels Numbers · First number 5 · Step 5 · Digits 2 · Placement radial, auto-flip · Size 0.9 · Font Futura Medium ☐
Chapter band	Radius 14.7 · Band width 2.6 ☒
Crosshair ticks	Radius 12 · Count 4 · Length 12 · Width 0.1 ☒
Sectors band	Radius 6 · Count 12 · Every 2 · Band width 6 ☐
Hour hand	Shape dauphine · Length 7.5 · Width 1.3 · Tail 1 · at 22 % ☒
Minute hand	Shape dauphine · Length 12.6 · Width 1.1 · Tail 1.4 · at 18 % ☒
Seconds hand	Shape lollipop · Length 13.4 · Width 0.12 · Tail 3 · Head · ring · disc 0.9 · at 80 % ☒
Cap	Ø 1.6 ☒

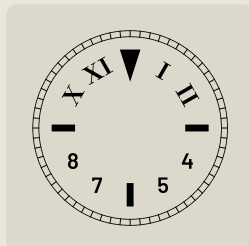
RECIPES

Railroad track	Ticks, Count 60. Two thin bands: one at the ticks' Radius, one at Radius – tick Length + Band width. (California, p. 10: 14.2 and 13.32.)
Fifths of a second	Ticks Count 60 above ticks Count 300; the 300 Skip where: all marks above.
Cardinals only	Every 3. The other eight hours: Every 3 + Skip those instead.
Minute numerals 05-60	Numerals, Labels Numbers, First number 5, Step 5, Digits 2, Start 30°.
24-hour ring	Numerals, Labels Custom: 24, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23. (Field watch, p. 7.)
Wordmark	Numerals, Count 1, Start 180° (or 0°), Labels Custom, Placement on path. (Dress watch, p. 8.)
Two fonts, one ring of hours	Two numerals rings at the same Radius, each with a Custom list that leaves the other's positions empty. (California, p. 10.)
A gap round numerals	A band below the numerals; Knock out: markers & numerals above; Clearance 0.3.
Sector dial	A band, Count 12, Every 2, Band width = Radius: alternate pies.
Lume opening	Hollow wall on a marker or a hand: 0.15–0.25.

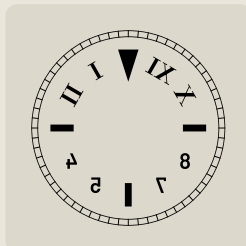
OUT the Export section



design
PDF · SVG · PNG



production
one colour, no plate, no hands



production + mirror
for toner transfer

PDF	vector, true size – Illustrator, print, the press
SVG	vector, true size – Illustrator, laser, the web
PNG	at the dpi you set – film for toner transfer
MP4 · GIF	the hands moving – Clip seconds, Frames per s
Live text	numerals stay text in PDF and SVG. Off by default: outlines need no fonts and print as seen
Settings	Save... and Load... a .json of everything; Defaults starts over

IN SVG · PDF · AI, for markers and hands

- Draw pointing up, at 12 o'clock. The artboard's centre is the anchor: a hand's pivot, a marker's centre.
- Black is ink. White shapes cut holes. Strokes become outlines.
- Outline text first. Images and gradients are skipped, and the log says so.
- Save AI files with "Create PDF Compatible File".

BEFORE YOU ETCH OR PRINT

- Draw at true size. A 30 mm dial at 600 dpi is 709 pixels across.
- Check the thinnest line at 100 %: a 0.09 mm tick is two pixels at 600 dpi.
- **Production** gives one colour: no plate, no hands, the date aperture as a cut line.
- **Mirror** flips it for toner transfer.

QUESTIONS

Settings?	Saved as you work, in <code>~/Library/Application Support/DialTool/last.json</code> .
Diameter?	Changing it moves nothing else: radii and lengths are mm from the centre.
A ring is gone?	Check its checkbox, Every and Offset, and what it skips.

Fonts?	Any font installed on the Mac.
Slow?	Install skia-pathops (p. 2). The log then says shape combining: skia-pathops .
Start over?	Export → Settings → Defaults.

WHEN SOMETHING GOES WRONG

1. Open Export and read the log. At launch it says **preview: canvas** and **all controls connected**.
2. If the preview looks wrong, switch Canvas → PDF and compare.
3. Send a screenshot and the log. If the window doesn't build at all, run `tools/dial_probe.py` and send its report.

WHERE THINGS LIVE

Code · docs	<code>github.com/duztman/dial-tool</code>
Words	<code>docs/glossary.md</code>
Presets	<code>docs/guide/presets/*.json</code>
This guide	<code>docs/guide/guide.html</code> → <code>tools/make_guide.py</code>